

NFL 1st Down Analysis – Assignment 3, Exercise 2

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- **Exploratory Data Analysis:**

Get the NFL data from the R library (nflfastR). Filter the data for regular season games since 2018, regular season only and start to explore the data. Do the following and anything else that helps you understand the data further.

- Find completion percentage for quarterbacks (passers) by season. Who had the highest, the lowest? (Min 100 passes)

season <int>	passer <chr>	pass_attempts <int>	completions <dbl>
2018	D.Brees	589	418
2019	D.Brees	422	307
2020	D.Brees	475	322
2021	T.Brady	838	544
2022	P.Mahomes	781	508
2023	K.Cousins	326	216

season <int>	passer <chr>	pass_attempts <int>	completions <dbl>
2018	J.Allen	349	169
2019	D.Haskins	234	119
2020	J.Hurts	161	77
2021	Z.Wilson	429	213
2022	S.Thompson	161	78
2023	D.Thompson-Robinson	117	60

- Find total rushing yards and average yards per attempt for all rushers by season. Who had the highest and lowest rushing yards per attempt (min 100 rushes)

season <int>	rusher <chr>	total_yards <dbl>	attempts <int>
2018	NA	6205	821
2019	NA	5986	835
2020	NA	6483	906
2021	NA	7174	953
2022	NA	7106	959
2023	D.Achane	809	109

season <int>	rusher <chr>	total_yards <dbl>	attempts <int>
2018	L.Blount	418	155
2019	P.Barber	472	155
2020	J.Kelley	354	111
2021	M.Ingram	554	160
2022	L.Fournette	679	194
2023	J.Hurts	283	110

Explore first down probability when a team is leading, trailing or tied in specific quarters.

qtr <dbl>	team_situation <chr>	total_plays <int>	first_down_plays <dbl>	first_down_probability <dbl>
1	Leading	2451	2451	100
1	Tied	7930	7930	100
1	Trailing	4428	4428	100
2	Leading	7152	7152	100
2	Tied	2426	2426	100
2	Trailing	8908	8908	100
3	Leading	6414	6414	100
3	Tied	993	993	100
3	Trailing	7728	7728	100
4	Leading	5747	5747	100
4	Tied	975	975	100
4	Trailing	10993	10993	100
5	Tied	363	363	100
5	Trailing	52	52	100

- For the 2023 season, examine how often the Super Bowl quarterbacks, Brock Purdy and Patrick Mahomes, converted 1st downs on their passes, on 1st, 2nd, 3rd and 4th down.

passer <chr>	do... <dbl>	num_plays <int>	num_first_downs <int>	conversion_rate <dbl>
B.Purdy	1	257	100	38.91051
B.Purdy	2	207	81	39.13043
B.Purdy	3	163	76	46.62577
B.Purdy	4	5	4	80.00000
P.Mahomes	1	324	NA	NA
P.Mahomes	2	317	NA	NA
P.Mahomes	3	235	99	42.12766
P.Mahomes	4	17	9	52.94118

- Examine the leading rushers for both teams, Isiah Pacheco and Christian McCaffrey and how often they converted runs into 1st downs on 1st, 2nd, 3rd, and 4th

rusher <chr>	down <dbl>	num_plays <int>	num_first_downs <int>
C.McCaffrey	1	197	32
C.McCaffrey	2	117	NA
C.McCaffrey	3	26	16
C.McCaffrey	4	3	2
I.Pacheco	1	190	27
I.Pacheco	2	82	33
I.Pacheco	3	17	10
I.Pacheco	4	3	2
I.Pacheco	NA	1	0

- Predictive Model / Probabilistic Machine Learning**

Build a predictive model incorporating factors such as field position, down and distance, and time remaining to forecast the probability of a first down. Be sure to filter your data for Regular Season games only (i.e., season_type = "REG") and play_type = "run" or "pass".

The nflfastr library has more than 350 columns and feel free to use any as your independent variables; however, the [Useful Categories](#)

[Download Useful Categories](#)

sheet on Canvas will help filter your data. A few different versions of the dependent variables can be used depending on how you set up your model.

first_down is a binary indicator if the play ended in a first down.

first_down_rush is a binary indicator if a running play converted the first down.

`first_down_pass` is a binary indicator if a passing play converted the first down.

- Should you run or pass?

Use your model to analyze various situations on behalf of your coach. Answer the questions below and write a summary considering that the audience is a football head coach (ie, non-technical). Filter the data on the play types “run” or “pass” and make a recommendation on which is preferable, running or passing...

For this model, I have included all the variables that the professor suggested. Logistic regression model was performed as it is used for variables that are binary. We could have made the model better by first doing regression model with all the variables and then removing others but that could be an improvement in the later ones. There are some missing values and we should not omit them. Rather if we did KNN analysis we could find those missing values. This model was not overfitted and does not have high-dimensionality but can be majorly improved. Variable importance was also not performed.

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Call:
glm(formula = first_down ~ ., family = binomial, data = training_data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-0.3395 -0.1113 -0.1058  0.0000  3.3203

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -5.955e+00  3.774e-01 -15.780  < 2e-16 ***
first_down_rush  2.705e+01  5.862e+02  0.046  0.96319
first_down_pass  2.687e+01  4.492e+02  0.060  0.95231
down           1.930e-02  1.235e-01  0.156  0.87583
ydstogo        7.507e-02  2.199e-02  3.414  0.00064 ***
game_seconds_remaining 4.837e-05  9.459e-05  0.511  0.60909
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 28346.4  on 23683  degrees of freedom
Residual deviance: 1247.2  on 23678  degrees of freedom
(86 observations deleted due to missingness)
AIC: 1259.2

Number of Fisher Scoring iterations: 20
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first_down <dbl>	first_down_rush <dbl>	first_down_pass <dbl>	down <dbl>	ydstogo <dbl>	game_seconds_remaining <dbl>	predicted_prob <dbl>
0	0	0	1	10	3600	0.006624206
0	0	0	2	7	3570	0.005390334
1	1	0	3	1	3535	0.999999999
0	0	0	1	10	3496	0.006591187
1	0	1	2	10	3492	1.000000000
0	0	0	1	10	3454	0.006577899
0	0	0	2	9	3416	0.006211935
0	0	0	3	9	3411	0.006330682
0	0	0	1	10	3359	0.006547940
0	0	0	2	11	3328	0.007180587
1	0	1	3	5	3291	1.000000000
0	0	0	1	10	3252	0.006514359
0	0	0	2	10	3248	0.006639168
0	0	0	3	13	3208	0.008446384

Outside of the red zone area (more than 20 yards from the goal line), do you prefer to run or pass?

- 3rd and short (3 or less yards to go) **PASS**

first_down <dbl>	first_down_rush <dbl>	first_down_pass <dbl>	down <dbl>	ydstogo <dbl>	game_seconds_remaining <dbl>	predicted_prob <dbl>	decision <chr>
0	0	0	3	3	3029	0.004396397	rush
1	1	0	3	1	2214	0.999999999	pass
0	0	0	3	1	1824	0.004196602	rush
0	0	0	3	3	1734	0.004649107	rush
1	0	1	3	3	1545	0.999999999	pass
1	0	1	3	3	986	0.999999999	pass

- 3rd and long (7 or more yards to go) **RUSH**

first_down <dbl>	first_down_rush <dbl>	first_down_pass <dbl>	down <dbl>	ydstogo <dbl>	game_seconds_remaining <dbl>	predicted_prob <dbl>	decision <chr>
0	0	0	3	10	3522	0.006074257	rush
0	0	0	3	16	3269	0.008245761	rush
0	0	0	3	16	3119	0.008299113	rush
0	0	0	3	12	2929	0.006875163	rush
0	0	0	3	9	2766	0.005974256	rush
1	0	1	3	10	2323	1.000000000	pass

- 4th and short (3 or less yards to go) **PASS**

first_down <dbl>	first_down_rush <dbl>	first_down_pass <dbl>	down <dbl>	ydstogo <dbl>	game_seconds_remaining <dbl>	predicted_prob <dbl>	decision <chr>
1	1	0	4	1	2814	0.999999999	pass
0	0	0	4	1	1873	0.004187378	rush
0	0	0	4	1	1105	0.004328535	rush
1	0	1	4	1	2636	0.999999999	pass
1	0	1	4	3	3258	0.999999999	pass
1	1	0	4	1	2613	0.999999999	pass

- 4th and long (7 or more yards to go) **RUSH**

A tibble: 6 × 8

first_down <dbl>	first_down_rush <dbl>	first_down_pass <dbl>	down <dbl>	ydstogo <dbl>	game_seconds_remaining <dbl>	predicted_prob <dbl>	decision <chr>
0	0	0	4	10	75	0.005922908	rush
0	0	0	4	15	22	0.008575785	rush
0	0	0	4	8	902	0.005308482	rush
0	0	0	4	14	196	0.008027247	rush
0	0	0	4	10	1803	0.006435905	rush
0	0	0	4	25	129	0.018086758	rush

Red Zone / Goal Line

- For the four situations above, how does your answer change in the Red Zone (Between 4 and 20 yards from the goal line)? How about 3rd and short at the goal line (three or less yards from the goal line)? Does your answer change for 4th and short at the goal line?

The model suggests PASS when there are 3 or fewer yards to go and RUSH when there are 7 yards or more. This is the case for both 3rd and 4th. From my understanding it is better to run it when the distance is greater and pass it when the distance is short. I do not have a lot of knowledge on football but this seems like a normal thing.

2024 NFC Championship Game Analysis

In the 2024 NFC Championship game, the Detroit Lions faced fourth-and-2, 28 yards from the San Francisco goal line while leading by 14 points with 7:03 left in the third quarter. Instead of attempting a 46-yard field goal, they failed on a pass. Based on Seth Walder's tweet, the Lions expected winning percentage was 0.2% points higher going for it

Later in the game, Detroit faced a fourth-and-3 30 yards from the San Francisco goal line. They were down by three points with 7:38 left in the fourth quarter. Instead of trying a 48-yard field goal to tie the game, the Lions failed on another pass conversion. Based on Seth Walder's tweet, the Lions expected winning percentage was 0.3% points higher going for it

The Lions ended up losing the game. Do you agree with what Detroit did, or do you think they should have kicked field goals in one or both situations? Write a brief summary of your thoughts. Feel free to cite research articles or any other publicly available information to answer this question (i.e., you don't have to build a model to answer this).

While stats and models can give us some idea of what might work best, they cannot fully predict the outcomes and it might not be what we want. Especially in close games like this one, there are a lot of unpredictable things happening on the field.

Sometimes it's just like predicting the weather. It says we might get rain tomorrow but because some factors have not been accounted for like wind, pattern changes, and temperature changes, it ends up raining. Similarly, in football, there are tons of variables that can affect the outcome of a play or a game. Even if a statistical model suggests one decision is better, there could still be things it didn't consider, like how tired the players are or if there's a specific strategy the other team is using.

So, while it's easy to look back and say what the Lions should have done, it's important to remember that football is a fast-paced, dynamic game with lots of moving parts. Sometimes these models miss things (could be even high-dimensionality, overfitting, or missing variables) and sometimes the coaches miss things.